

# Neoliberal conservation in REDD+: The roles of market power and incentive designs

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## ABSTRACT

Market power relations and incentive designs are critical determinants affecting the success or failure of implementing the scheme of Reducing Emissions from Deforestation and Degradation (REDD+). This paper explores how market power differences in REDD+ and incentive designs with different motivating objects affect the actors' own emissions-reducing behavior. Guided by a theoretical framework that considers both market power and incentive designs, this paper examines the impacts of incentive designs and market power on the behavior of actors in REDD+ by constructing game models. A numerical simulation is applied to examine the effects of parameter changes on the optimal strategies of each actor under different market-power structures and different incentive solutions. The results demonstrate the following. First, the incentives for landholders in REDD+ are more conducive to affect actors' efforts to reduce emissions than the incentives for the investor. Second, landholders, rather than investors, were found to obtain more market power due to the unequal market-power structure in REDD+, which will ultimately improve the landholders' efforts to conserve forests and make a profit. At the same time, investors' efforts and profits were also found not decline, despite the existence of the unequal market power. Third, the effects of market power and incentive designs on actors were found to be related to various REDD+ factors, including opportunity costs, payments, the price of carbon credits, and the output elasticities of efforts. This paper further demonstrates that the incentive designs and market power have a more significant impact on landholders than on investors.

## 1. Introduction

Emissions from deforestation and forest degradation have become the target of global environmental governance, as they are one of the primary sources of anthropogenic greenhouse gas emissions (van der Werf et al., 2009). There is an urgent need for action to conserve forests, in order to link regional interests with regional capacities and thus help to mitigate global climate change (Salles et al., 2017). In light of this, the Reducing Emissions from Deforestation and Degradation (REDD+) programme was proposed at the 13<sup>th</sup> session of the Conference of the Parties (COP 13) in 2007 to provide incentives for developing countries to reduce emissions from the forestry sector (UNFCCC, 2008). After eight years of discussion, REDD+ was incorporated into the Paris Agreement at the 21<sup>st</sup> session of the Conference of the Parties (COP 21) in 2015, to replace the Kyoto Protocol that expires in 2020 (UNFCCC, 2016). REDD+ is considered to be a

low-cost option for mitigating climate change compared with other programs, such as renewable energy schemes (Rakatama et al., 2017).

The REDD+, as an application of payments for ecosystem services (PES) in forest conservation, aims to reward developing countries for avoiding deforestation and forest degradation to reduce carbon emissions (Karsenty et al., 2014). The theoretical basis of PES is considered to be based on the Rational Choice Theory and the Coase theorem (Kolinjivadi et al., 2019). Thus, the landholders are assumed to participate rationally in the PES and adopt environmentally friendly behavior if the incentives reflecting the value of the beneficiaries of particular ecosystem services can compensate or exceed their opportunity costs (Pagiola and Platais, 2007). The originality of REDD+ lies in the incentive mechanism that provides individuals, communities, and states with financial incentives to conserve forests and increase carbon sinks (Karsenty and Ongolo, 2012). Proponents of neoliberalism argue that since the market is more efficient than the policy-makers in allocating

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scarce resources, natural resources are most likely to be conserved when they are assigned economic value through market mechanisms (Humphreys, 2009). In line with this, REDD + practices transform avoided deforestation into a new tradable commodity that can be traded on the international carbon market or purchased with government funds (Fletcher et al., 2017). REDD + follows the principles of neoliberal environmentalism that employs market-based instruments (MBIs) rather than “command-and-control” policies to achieve sustainable forest management (Büscher et al., 2014). Neoliberal environmentalism adopts market mechanisms as the primary instruments to conduct individual behavior. In this context, technique and rationality based on economic growth, cost-benefit analysis, and financial incentives are applied to encourage and reward conservation (Fletcher, 2010). Thus, mobilizing market rationality represents an external environmental intervention rather than an internal individual belief. Indeed, the process of REDD + implementation is itself a process of constant game and negotiation among investors, landholders and policy-makers. As such, REDD + being based on a PES scheme can be regarded as an application of neoliberal conservation in the international forestry sector (Fletcher and Büscher, 2017).

Neoliberal conservation often involves new forms of market power, “particularly those that seek to re-shape local subjectivities by both conservationist and neoliberal-economic values” (Holmes and Cavanagh, 2016). Although current debates on REDD + focus mainly on purely technical aspects, REDD + itself also involves different types and dimensions of market power. REDD + transfers funds from carbon credit buyers to landholders, the payments being conditional on the sustainable land use and the reduction of emissions according to set baselines (Corbera, 2012). This scheme thus links local forest conservation with the demands of industrialized countries, affecting market-power relationships among the relevant actors in various ways. Due to owning more information about the opportunity costs for providing ecosystem services than investors, landholders can turn private information into market power to obtain information rents from investors (Ferraro, 2008). In addition, those landholders whose income comes from off-farm activities are reluctant to provide ecosystem services due to the anticipation of urban sprawl, placing them in dominant negotiating position with buyers. Kosoy and Corbera (2010) argued that the asymmetries of market power characterize the process of production, exchange, and consumption of ecosystem services. Due to complicated market-power struggles between actors, many countries have encountered challenges in the early stages of implementing REDD + (Korhonen-Kurki et al., 2014). It is, therefore, necessary to examine the status of participants in policy interactions and the market-power relationships to ensure the successful implementation of REDD + (Brockhaus et al., 2014).

While market power is often emphasized as a critical factor in forest policy research, to date, there has been scant literature concerning the role of market power in REDD + (Krott et al., 2014). Van Hecken et al. (2015) argued that PES transactions conceal more profound asymmetries of market power that may reproduce or exacerbate inequalities in the acquisition and use of natural resources. Makatta et al. (2015) found that there were imbalances of market power and struggles in the Participatory Forest Management (PFM) projects, which hindered the development of the REDD + programme in Tanzania. Based on a unique set of interviews and raw data analysis, Rowe (2015) examined the market-power relationships active in REDD + and how these shaped the programme's outcomes. Bouthavong et al. (2017) investigated the participation of actors in the three REDD + pilot projects in Laos, finding that the distribution of decision-making power was imbalanced and mainly concentrated in the central government and partner development organizations. Similarly, Patel et al. (2013) found that local communities have more market power than other actors to determine the allocation of REDD + funds in a REDD + pilot project in Nepal. Although many studies make the argument that the market power of actors in REDD + should be balanced, Brockhaus and Di Gregorio

(2014) argued that “although cooperation is perceived as ideal in any collective decision-making setting, a certain level of conflict is necessary for progress in REDD + decision-making.” Bearing this argument in mind, the impacts of market power on REDD + have not yet become clear. In the current paper, therefore, we are particularly interested in examining the role of the different market-power relationships in REDD +.

In addition, incentive design is also a key role in REDD +. The incentive in REDD + as a market-oriented policy tool can internalize the social costs of the REDD + mechanism to be compatible with social and private benefits (Nasi et al., 2011). As a result, incentive solutions are increasingly seeking more effective results by bundling the multiple types of costs for providing ecosystem services (Bryan, 2013). REDD + incentives often take many forms, including monetary compensation paid directly to individuals (Kaczan et al., 2013), in-kind payments (Sommerville et al., 2010), payments for labor exchange (Turpie et al., 2008), and informal contractual agreements based on collective resource-management principles (Escobar et al., 2013; Miao et al., 2019). Although these methods specifically identify the target effect of incentives on behavior in REDD +, they do not systematically examine the impacts of incentive designs with different motivating objects on the behavior of various actors. Indeed, REDD + involves multiple actors, and the impacts of different incentive designs on actors' behavior are heterogeneous (Sheng et al., 2016). The difference in market power between actors makes the impact of incentive designs more complicated. Therefore, it is necessary to incorporate market power and incentive designs into a new theoretical framework and analyze the roles of the two on the behavior of various actors.

Market power has been construed as having diverse types and dimensions in social sciences. For this paper, market power refers to the ability of actors to use their market positions to influence forestry and land-use decision-making so that the outcomes of these decision-making processes can serve their interests (Biermann, 2010; Krott et al., 2014). The actors in REDD + include investors (enterprises, NGOs, and financial institutions), landholders and policy-makers (Rantala and Di Gregorio, 2014). These actors each seek discursive power in the design and implementation of REDD + in order to support their interests (Schroeder, 2010). In the design of REDD +, the policy-makers as an intermediary can provide investors and landholders with economic incentives to motivate them to reduce carbon emissions. Thus, a critical issue of REDD + incentive design is who should receive such incentive payments from states. Different incentive designs on this issue have had different effects on landholders and investors through specific incentive policies, which have, in turn, affected REDD + project performance. In the implementation of REDD +, both landholders and investors use their knowledge, material wealth, opportunity costs and other private information as a source of market power, striving to dominate the decision-making process (Ferraro, 2008; Rowe, 2015). Therefore, both in terms of REDD + incentive design and implementation, there exists market power affecting the behavior of landholders and investors.

While neoliberal policies have been widely applied in the studies of forest conservation, to date there has been a lack of quantitative research on how market power and incentive solutions interact and lead to political-ecological outcomes (Winkel, 2012). In response to this gap, this paper extends the current analyses from market-power perspectives to identify the roles of incentive designs and market power in REDD +. Specifically, we examine the impacts of different incentive solutions and market-power relationships on the actors' behavior in REDD + in order to postulate three arguments. First, the incentives to landholders in REDD + are more conducive to encouraging proper emissions-reducing conduct than the incentives to investors. Second, landholders, rather than investors, can obtain more market power due to the unequal market-power structure in REDD +, which will ultimately improve the landholders' efforts to conserve forests and make a profit. Third, incentive designs and market power have a more significant impact on landholders than investors.

The remainder of this paper is organized as follows. Section two establishes game models to describe how landholders and investors choose their optimal strategies under the combined effect of two incentive designs (landholder- and investor-incentive solutions) and two market-power relationships (equal and unequal market power). Section three presents the results of the numerical simulation to illustrate changes in actor efforts and benefits to reduce emissions that are affected by incentive designs and market power. Section four discusses the results, and section five presents our conclusions.

## 2. Methods

### 2.1. Model specification

The major actors in REDD + projects include landholders, investors, and policy-makers. The landholders are the providers of forest ecosystem services, while the investors are the buyers. To simplify the analysis, this study treats many landholders and investors as two groups, regardless of the differences between individuals within the group. Although this assumption ignores the impact of the pluralistic values among individuals on actor behavior, private investment may lose interest when REDD + contains pluralistic values (Wunder, 2005). Due to the declining public sector resources, the lack of private investment will make the REDD + projects even more impossible, leading to the pluralistic values not being realized in REDD + (Fletcher and Büscher, 2017). Therefore, we treat all investors and landholders as two homogeneous groups to examine the impacts of market power and incentive designs on the behavior of these two groups.

The landholders receive payments from the investors for reducing emissions from deforestation. The payment amount depends on the number of reduced emissions ( $q$ ) and the payment per unit of reduced emissions ( $r$ ). Landholders participating in REDD + projects are required to give up the income from the original land use. The abandoned benefits become opportunity costs, depending on the opportunity costs per unit of reduced emissions ( $c_o$ ) and the number of reduced emissions ( $q$ ). Moreover, the landholders are required to make efforts for forest conservation, such as investing in infrastructure, labor and machinery, which constitute implementation costs in the REDD + programme (Rakatama et al., 2017). Following Sheng and Qiu (2018), we assume that landholder implementation costs ( $c_L$ ) are only related to the latter's efforts ( $e_L \in (0, +\infty)$ ), and will increase as efforts rise (i.e.,  $c_L' > 0$ ). In addition, we assume that there occurs an increase in marginal costs to reduce emissions (i.e.,  $c_L'' > 0$ ). According to Cvitanic et al. (2008), the landholder implementation cost can be expressed as a quadratic function:

$$c_L = \frac{1}{2} \theta_L e_L^2 \quad (1)$$

Where  $\theta_L \in (0, +\infty)$  represents the landholder cost coefficient of reduced emissions. The investors can obtain "carbon credits" by investing in REDD + projects, and then earn revenue by selling on these carbon credits in the international carbon markets (Cheng et al., 2018). This revenue is determined by the number of reduced emissions ( $q$ ) and the price of carbon credits ( $p$ ). Investor costs in reducing emissions mainly consist of the payment per unit of reduced emissions ( $r$ ) and transaction costs ( $c_D$ ). The transaction costs are derived from establishing MRV (monitoring, reporting, and verification) systems, and the negotiation and management costs of REDD + contracts (Gregersen et al., 2010; Merger et al., 2012) (Gregersen, 2010 #4; Karky, 2010 #6; Merger, 2012 #104). Similarly, investor costs in reducing emissions ( $c_D$ ) are assumed to be an increasing function of their efforts. whereby ( $e_D \in (0, +\infty)$ ), and  $c_D' > 0$ ,  $c_D'' > 0$ . The investor cost function is set as follows:

$$c_D = \frac{1}{2} \theta_D e_D^2 \quad (2)$$

Where  $\theta_D \in (0, +\infty)$  represents the investor cost coefficient of reduced

emissions in REDD +. For the sake of simplicity, we assumed that the landholders and investors have the same cost coefficient of reduced emissions, i.e.,  $\theta_L = \theta_D = \theta$ .

We also assumed that the amount of reduced emissions ( $q$ ) in REDD + is related to landholders and investors' efforts. In line with Nishitani et al. (2011), a Cobb-Douglas function is applied to establish the functional relationship between the number of reduced emissions and actors' efforts:

$$q = A e_L^\alpha e_D^\beta \quad (3)$$

Where  $q$  represents the number of reduced emissions;  $e_L$  and  $e_D$  represent landholder and investor efforts to reduce emissions in REDD +, respectively; and  $A \in (0, +\infty)$  represents the total factor productivity of reduced carbon emissions. The parameters  $\alpha \in (0, 1)$  and  $\beta \in (0, 1)$  represent the output elasticity of landholder and investor efforts, respectively;  $0 < \alpha + \beta < 1$ , referring to the number of reduced emissions in REDD +, is a function of decreasing returns to scale (Nishitani et al., 2011).

The policy-makers, as another critical actor in REDD +, can establish incentive policies affecting the distribution of benefits between landholders and investors. Indeed, the role of the policy-makers in REDD + remains central to what is possible and what is valid, and REDD + cannot be completed without the participation of the policy-makers (Rodriguez-Ward et al., 2018). REDD + projects can generate non-carbon benefits, such as protecting biodiversity and improving the welfare of the whole society (Ojea et al., 2016). However, these external benefits have not been included in the price of carbon credits and the payments from the investors to the landholders. Therefore, it is necessary for policy-makers to provide incentives to landholders and investors to encourage them to participate in REDD +.

We assumed that landholders and investors are the two key players in the game, while the policy-makers act as independent actors deciding upon incentive policies. All of them were assumed to be risk-neutral and rational profit-maximizing subjects (Fletcher, 2010). However, it is difficult for the policy-makers to have an accurate insight into the costs borne by landholders and investors in reducing emissions. Thus, the policy-makers can only formulate incentive policies according to the number of reduced emissions ( $q$ ), while landholders and investors can make decisions according to the incentive policies. To examine the effects of different incentive designs and market-power relations on the decision-making of both the latter actors, we proposed two incentive solutions: the landholder-incentive solution and the investor-incentive solution. The incentives per unit of reduced emissions are set as the same in both incentive solutions.

### 2.2. Strategies under the landholder-incentive solution

The landholders abandon the income from other possible land uses when participating in REDD +. Thus, the policy-makers should fully compensate this with external benefits, in order to encourage landholders to provide ecosystem services. We assumed that the policy-makers only provides incentives per unit of reduced emissions ( $s$ ) for landholders. The incentive for landholders ( $s$ ) was set as an exogenous variable. Landholder and investor profits can be formulated as follows:

$$\pi_L^1 = (r - c_o + s)q - c_L = (r - c_o + s)A e_L^\alpha e_D^\beta - \frac{\theta e_L^2}{2} \quad (4)$$

$$\pi_D^1 = (p - r)q - c_D = (p - r)A e_L^\alpha e_D^\beta - \frac{\theta e_D^2}{2} \quad (5)$$

Where  $\pi_L^1$  and  $\pi_D^1$  represent landholder and investor profits under the landholder-incentive solution, respectively;  $s \in [0, 1]$  represents landholder incentive per unit of reduced emissions. According to Eqs. (4) and (5), the profits ( $\pi_L^1$  and  $\pi_D^1$ ) are the concave functions of their respective efforts ( $e_L$  and  $e_D$ ), i.e.,  $\partial^2 \pi_L^1 / \partial e_L^2 < 0$ ,  $\partial^2 \pi_D^1 / \partial e_D^2 < 0$ . This means that each game model has a unique optimal solution. Next, we

classified market-power relationships into two market-power structures of equality and inequality, and then examined the decision-making of both actors under each of these market-power structures.

### 2.2.1. Strategies in an equal market-power structure

The landholders and investors can make decisions concurrently when they have equal market power. Given the assumption of profit-maximizing subjects, the optimal efforts of landholders and investors can be obtained by solving the Nash game model (as shown in Appendix A.1):

$$\begin{cases} e_{L1}^1 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p-r))^{\frac{\beta}{2(2-\alpha-\beta)}} (\alpha(r-co+s))^{\frac{2-\beta}{2(2-\alpha-\beta)}} \\ e_{D1}^1 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p-r))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} (\alpha(r-co+s))^{\frac{\alpha}{2(2-\alpha-\beta)}} \end{cases} \quad (6)$$

Where  $e_{L1}^1$  and  $e_{D1}^1$  represent the optimal efforts of landholders and investors in an equal market-power structure, respectively. According to Eqs. (4), (5) and (6), the optimal profits of landholders ( $\pi_{L1}^1$ ) and investors ( $\pi_{D1}^1$ ) are as follows:

$$\begin{cases} \pi_{L1}^1 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r-co+s))^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p-r))^{\frac{\beta}{2-\alpha-\beta}} \left(\alpha^{\frac{\alpha}{2-\alpha-\beta}} - \frac{1}{2}\right) \\ \pi_{D1}^1 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r-co+s))^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p-r))^{\frac{2-\alpha}{2-\alpha-\beta}} \left(\beta^{\frac{\beta}{2-\alpha-\beta}} - \frac{1}{2}\right) \end{cases} \quad (7)$$

Where  $\pi_{L1}^1$  and  $\pi_{D1}^1$  represent the optimal profits of landholders and investors in an equal market-power structure, respectively.

### 2.2.2. Strategies in an unequal market-power structure

When the market power is unequal, the positions of landholders and investors are asymmetric, resulting in a sequence of decisions. With more information about the opportunity costs of providing ecosystem services than investors, landholders can turn private information into market power to obtain information rents (Ferraro, 2008). Thus, the landholders are the leaders who have priority in terms of decision-making regarding optimal efforts to maximize their profits (Sheng et al., 2016). The investors, as the follower, then determine their optimal efforts based on the landholders' strategy and public incentives. The Stackelberg game model can be employed to examine the game process of landholders and investors conduct in the context of an unequal market-power structure. Thus, the optimal efforts of landholders and investors can be obtained by solving the Stackelberg game model (as shown in Appendix A.2):

$$\begin{cases} e_{L2}^1 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} \left(\frac{2\alpha}{2-\beta}\right)^{\frac{2-\beta}{2-\alpha-\beta}} (r-co+s)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (\beta(p-r))^{\frac{\beta}{2(2-\alpha-\beta)}} \\ e_{D2}^1 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} \left(\frac{2\alpha}{2-\beta}\right)^{\frac{\alpha}{2-\alpha-\beta}} (r-co+s)^{\frac{\alpha}{2(2-\alpha-\beta)}} (\beta(p-r))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} \end{cases} \quad (8)$$

Where  $e_{L2}^1$  and  $e_{D2}^1$  represent the optimal efforts of landholders and investors in an unequal market-power structure, respectively. According to Eqs. (4), (5) and (8), the optimal profits of the landholders ( $\pi_{L2}^1$ ) and investors ( $\pi_{D2}^1$ ) are as follows:

$$\begin{cases} \pi_{L2}^1 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left(\frac{2\alpha(r-co+s)}{2-\beta}\right)^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p-r))^{\frac{\beta}{2-\alpha-\beta}} \\ \quad \left(\left(\frac{2\alpha}{2-\beta}\right)^{\frac{\alpha}{2-\alpha-\beta}} - \frac{1}{2}\right) \\ \pi_{D2}^1 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left(\frac{2\alpha(r-co+s)}{2-\beta}\right)^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p-r))^{\frac{2-\alpha}{2-\alpha-\beta}} \left(\beta^{\frac{\beta}{2-\alpha-\beta}} - \frac{1}{2}\right) \end{cases} \quad (9)$$

Where  $\pi_{L2}^1$  and  $\pi_{D2}^1$  represent the optimal profits of landholders and investors in an unequal market-power structure, respectively.

### 2.3. Strategies under the investor-incentive solution

REDD + as an application of the neoliberalization of nature seeks to use MBIs to conserve forests and reduce emissions. However, the current incentives in REDD + tend to be "burdened" with numerous public concerns around "alleviating poverty, gender, indigenous people, human rights, and other noble causes" (Wunder, 2005). Thus, it is necessary to provide incentives for investors to ensure the programme's attractiveness to the private sector. We assumed that the policy-makers only provide incentives per unit of reduced emissions ( $s$ ) for the investors. Landholder and investor profits can be expressed as follows:

$$\pi_L^2 = (r-co)q - c_L = (r-co)Ae_L^\alpha e_D^\beta - \frac{\theta e_L^2}{2} \quad (10)$$

$$\pi_D^2 = (p-r+s)q - c_D = (p-r+s)Ae_L^\alpha e_D^\beta - \frac{\theta e_D^2}{2} \quad (11)$$

Where  $\pi_L^2$  and  $\pi_D^2$  represent the profits of the landholders and investors under the investor-incentive solution, respectively;  $s \in [0, 1]$  represents the investor incentive per unit of reduced emissions. Similar to the landholder-incentive solution, the profits ( $\pi_{L2}$  and  $\pi_{D2}$ ) under the investor-incentive solution are concave functions of their own efforts ( $e_L$  and  $e_D$ ), i.e.,  $\partial^2 \pi_{L2} / \partial e_{L2}^2 < 0$ ,  $\partial^2 \pi_{D2} / \partial e_{D2}^2 < 0$ . We also classified market power into two structures of equality and inequality.

#### 2.3.1. Strategies in an equal market-power structure

In the context of an equal market-power structure, the landholders and investors can choose their optimal strategies simultaneously. The optimal efforts of landholders and investors can be obtained by solving the Nash game model (as shown in Appendix B.1):

$$\begin{cases} e_{L1}^2 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p-r+s))^{\frac{\beta}{2(2-\alpha-\beta)}} (\alpha(r-co))^{\frac{2-\beta}{2(2-\alpha-\beta)}} \\ e_{D1}^2 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p-r+s))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} (\alpha(r-co))^{\frac{\alpha}{2(2-\alpha-\beta)}} \end{cases} \quad (12)$$

Where  $e_{L1}^2$  and  $e_{D1}^2$  represent the optimal efforts of landholders and investors in an equal market-power structure, respectively. According to Eqs. (10)–(12), the optimal profits of the landholders ( $\pi_{L1}^2$ ) and investors ( $\pi_{D1}^2$ ) can be expressed as follows:



**Table 1**  
The numerical specifications.

| Parameters | Definitions                                           | Unit            | Value |
|------------|-------------------------------------------------------|-----------------|-------|
| $p$        | Price of carbon credits                               | USD/ ton carbon | 6.02  |
| $r$        | payment per unit of reduced carbon emissions          | USD/ ton carbon | 3.50  |
| $c_o$      | Opportunity cost per unit of reduced emissions        | USD/ ton carbon | 2.50  |
| $s$        | Incentive per unit of reduced carbon emissions        | USD/ ton carbon | 0.50  |
| $\alpha$   | Output elasticity of landholder effort                |                 | 0.20  |
| $\beta$    | Output elasticity of investor effort                  |                 | 0.30  |
| $\theta$   | Cost coefficient of reduced carbon emissions          |                 | 2.00  |
| $A$        | Total factor productivity of reduced carbon emissions |                 | 8.00  |

$$\left\{ \begin{array}{l} \pi_{L1}^2 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r-co))^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p-r+s))^{\frac{\beta}{2-\alpha-\beta}} \left( \alpha^{\frac{2-\alpha}{2-\alpha-\beta}} - \frac{1}{2} \right) \\ \pi_{D1}^2 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r-co))^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p-r+s))^{\frac{2-\alpha}{2-\alpha-\beta}} \left( \beta^{\frac{\beta}{2-\alpha-\beta}} - \frac{1}{2} \right) \end{array} \right. \quad (13)$$

Where  $\pi_{L1}^2$  and  $\pi_{D1}^2$  represent the optimal profits of landholders and investors in an equal market-power structure, respectively.

### 2.3.2. Strategies in an unequal market-power structure

Since landholders with more market power in REDD + have the priority in terms of decision-making to determine the optimal efforts to enable them to maximize profits, the optimal efforts of landholders and investors can be obtained by solving the Stackelberg game model (as shown in Appendix B.2):

$$\left\{ \begin{array}{l} e_{L2}^2 = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\alpha-\beta}} \left( \frac{2\alpha}{2-\beta} \right)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (r-co)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (\beta(p-r+s))^{\frac{\beta}{2(2-\alpha-\beta)}} \\ e_{D2}^2 = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\alpha-\beta}} \left( \frac{2\alpha}{2-\beta} \right)^{\frac{\alpha}{2(2-\alpha-\beta)}} (r-co)^{\frac{\alpha}{2(2-\alpha-\beta)}} (\beta(p-r+s))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} \end{array} \right. \quad (14)$$

Where  $e_{L2}^2$  and  $e_{D2}^2$  represent the optimal efforts of landholders and investors in an unequal market-power structure, respectively. According to Eqs. (10), (11) and (14), the optimal profits of the landholders ( $\pi_{L2}^2$ ) and investors ( $\pi_{D2}^2$ ) can be formulated as follows:

$$\left\{ \begin{array}{l} \pi_{L2}^2 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left( \frac{2\alpha(r-co)}{2-\beta} \right)^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p-r+s))^{\frac{\beta}{2-\alpha-\beta}} \left( \left( \frac{2\alpha}{2-\beta} \right)^{\frac{\alpha}{2-\beta}} - \frac{1}{2} \right) \\ \pi_{D2}^2 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left( \frac{2\alpha(r-co)}{2-\beta} \right)^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p-r+s))^{\frac{2-\alpha}{2-\alpha-\beta}} \left( \beta^{\frac{\beta}{2-\alpha-\beta}} - \frac{1}{2} \right) \end{array} \right. \quad (15)$$

Where  $\pi_{L2}^2$  and  $\pi_{D2}^2$  represent the optimal profits of landholders and investors in an unequal market-power structure, respectively.

## 3. Numerical simulation

### 3.1. Simulation design

To explore the effects of different incentive designs and market-power structures on the strategies of both actors in REDD+, we

constructed numerical simulations to compare the optimal efforts and profits of landholders and investors. Since the previous models suggest that optimal efforts and profits are determined by various parameters, we needed to examine the game results in terms of different parameter variations. In line with the average price of carbon emissions futures in 2016 on Investing.com (Hong et al., 2017), we set the price of carbon credits as 6.02 USD/ton carbon. Following a review of 29 empirical studies on REDD + opportunity costs, Boucher (2008) argued that the average opportunity cost was 2.51 USD/ ton carbon. Moreover, Boucher (2008) also found that the generalized transaction cost, comprising the transaction cost, implementation cost and administrative cost, was approximately 1 USD/ ton carbon in REDD +. Thus, for the current study, we set the payment per unit of reduced emissions ( $r$ ) as 3.50 USD/ ton carbon, representing the sum of the opportunity cost and the generalized transaction cost. To compensate for the external benefits and ensure attractiveness for the private sector, the incentive per unit of reduced emissions was set as 0.5 USD/ ton carbon for landholders and investors. Since the range of the output elasticities for landholders and investors' efforts ( $\alpha$  and  $\beta$ ) were  $0 < \alpha < 1$ ,  $0 < \beta < 1$  and  $0 < \alpha + \beta < 1$ ,  $\alpha$  and  $\beta$  were set as 0.2 and 0.3, respectively, in line with the work of Sheng and Qiu (2018). Since  $\theta > 0$  and  $A > 0$ , the parameters  $\theta$  and  $A$  were set to two and eight, respectively, to ensure that all the profit functions made sense. Since parameters  $\theta$  and  $A$  did not have significant impacts on actors' behavior in REDD+ (Sheng and Qiu, 2018), these two parameters were not simulated in this study. Table 1 presents the numerical specifications.

### 3.2. Results

#### 3.2.1. Effects of government incentives

Following the method of Sheng et al. (2016), the incentives per unit of reduced carbon emissions ( $s$ ) were set as 0–1 USD/ton carbon. Fig. 1 presents the effects of incentive solutions and market power on landholders and investors' strategies when the incentive amount changes.

Fig. 1 shows the effects of the incentive solutions on the optimal strategies of the landholders and investors. According to Fig. 1, both the efforts and profits of the two actors are positively related to the incentives ( $s$ ). The landholders' efforts and profits rise more significantly under the landholder-incentive solution as the incentive amount increases, while the investors' efforts and profits rise faster under the investor-incentive solution. These results suggest that the actors experiencing an incentive solution in line with their role would be more willing to participate in REDD + as they could thus obtain more incentives and receive more revenue. Since the investors are assumed to have higher marginal profits and output elasticity than the landholders ( $p-r > r-c_o$  and  $\beta > \alpha$ ), the investors' efforts and profits are always higher than the landholders'.

Fig. 1 also presents the impacts of the market power on actors' behavior. Regardless of which incentive solution is at play, the landholders' efforts and profits are consistently subject to greater differences under the two market-power structures than those of the investors. This suggests that the influence of market power on the landholders is higher than that on the investors. The differences caused by different market-

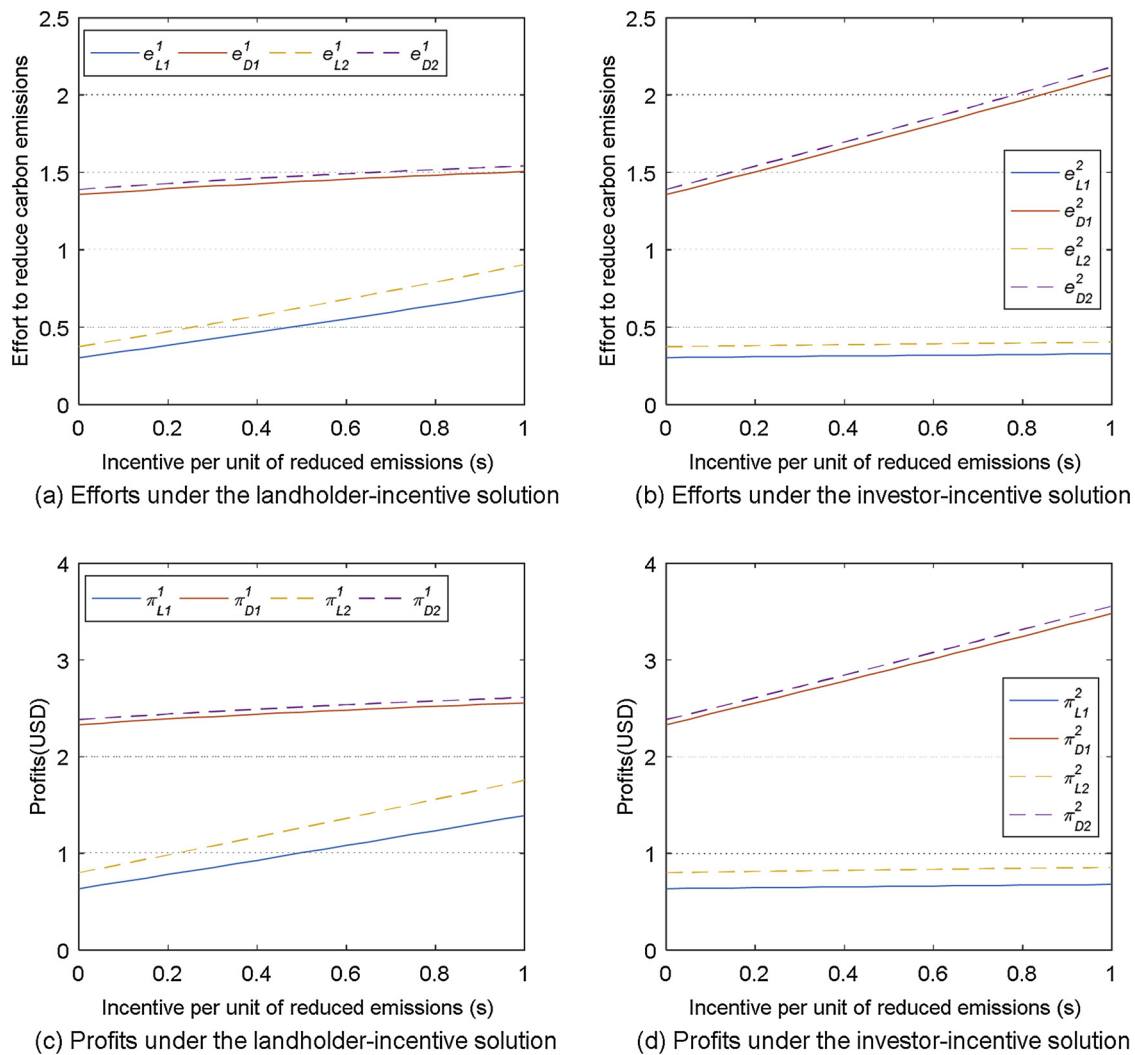


Fig. 1. The effects of incentive per unit of reduced carbon emissions ( $s$ ).

power structures increase more significantly with the increase in the incentive amount under the landholder-incentive solution, while these differences remain largely unchanged under the investor-incentive solution. Thus, landholders with more market power in decision-making can be assumed to be more inclined to make more significant efforts in forest conservation, in order to obtain more profit.

### 3.2.2. Effects of payment

Due to  $p > r > c_o > 0$ , the payment per unit of reduced carbon emissions ( $r$ ) was set as 2.6–5.6 USD/ton carbon. Fig. 2 presents the effects of incentive solutions and market power on landholders and investors' strategies when the payments change.

Fig. 2 shows the effects of the incentive solutions on the actors' optimal strategies. Landholders' efforts and profits are shown to rise continuously as payments ( $r$ ) increase, while those of investors decrease. According to Fig. 2(b) and (d), since the incentives received by the investors are much higher than the payments, under the investor-incentive solution the increase in payments will increase the investors' efforts and profits when the payments are small. Fig. 2(a) and (c) show that the landholders' efforts and profits will tend to stabilize when the payments increase to a certain level under the landholder-incentive solution. Furthermore, there is evidence of an equilibrium payment that equalizes the profits of the landholders and investors, meaning that the benefits are equitably distributed among the REDD + actors.

Fig. 2 also shows the impacts of the market power on actors'

behavior. The two actors' efforts and profits in the context of an unequal market-power structure emerge as greater than those in an equal internal power structure. Moreover, regardless of the incentive solution at play, landholders' efforts and profits consistently contain greater gaps under the two market-power structures than those of investors. This outcome indicates that the landholders are more sensitive to changes in the internal power structure as these actors play a leading role in REDD + . The gaps that arise in landholders' efforts and profits under the different market-power structures do so with an increase in payments; this trend is particularly significant under the landholder-incentive solution. However, the increase in payments reduces the effects of the market power on the investors' efforts and profits, since gaps in the latter continue to decrease.

### 3.2.3. Effects of opportunity cost

Due to  $r > c_o > 0$ , the opportunity costs ( $c_o$ ) were set as 0.5–3 USD/ton carbon. Fig. 3 presents the effects of incentive solutions and market power on landholders and investors' strategies when the opportunity costs change.

Fig. 3 presents the effects of the incentive solutions on actors' behavior. The two actors' efforts and profits emerge as inversely proportional to the opportunity costs; the two actors' optimal effort and profit changes under the same incentive solutions have similar trends regardless of the market-power structures. Moreover, the landholders' optimal efforts and profits are seen to decline faster than those of the

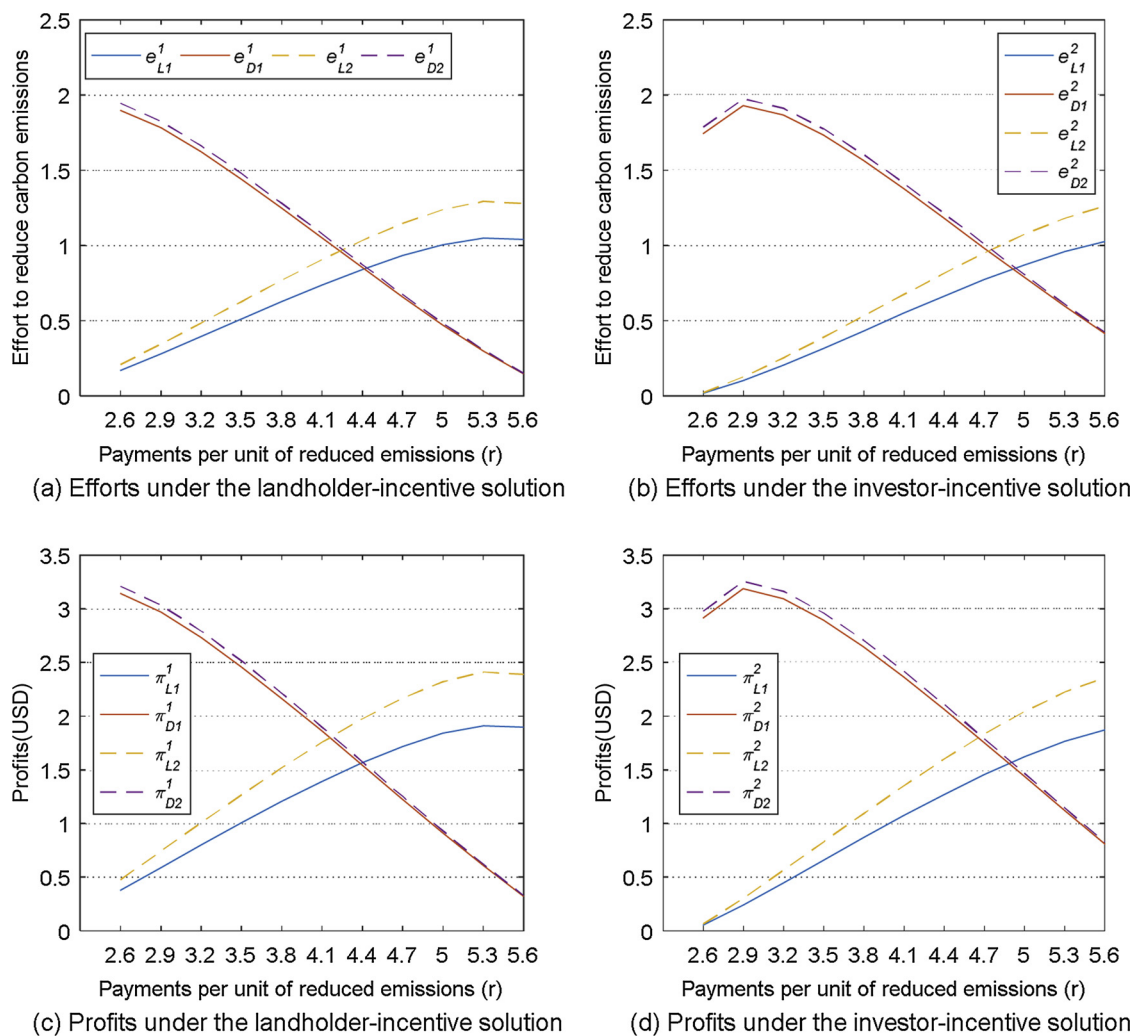


Fig. 2. The effects of payment per unit of reduced carbon emissions ( $r$ ).

investors with the increase in opportunity costs. This indicates that the landholders undergoing high opportunity costs will be reluctant to participate in REDD+, resulting in a reduction in investor profits.

Fig. 3 also presents the influence of the two studied market-power structures on actors' behavior. Regardless of the incentive solution at play, the efforts and profits of both actors in the context of an unequal market-power structure are shown to be higher than those under an equal market-power structure, especially for the landholders. Furthermore, the landholders' efforts and profits are consistently subject to greater differences under the two market-power structures than those of the investors. As opportunity costs rise, the effects of the market-power structures on the landholders' efforts and profits consistently diminish, while the differences in the investors' efforts and profits under the two power structures do not change significantly. This outcome suggests that when opportunity costs are not adequately compensated for, the landholders will not make further efforts to reduce emissions even if these actors have greater balances of market power in decision-making. Thus, while landholders with a greater degree of market power could benefit from a power imbalance, this advantage will diminish as opportunity costs increase.

### 3.2.4. Effects of the price of carbon credits

Due to  $p > r > 0$ , the price of carbon credits ( $p$ ) was set as 4–8 USD/ton carbon. Fig. 4 presents the effects of incentive solutions and market power on landholders and investors' strategies when the price changes.

Fig. 4 reveals the effects of the incentive solutions on actors' optimal

strategies. The two actors' efforts and profits can be seen to be proportional to the price of carbon credits ( $p$ ). The optimal effort and profit of each actor under the same incentive solution have similar trends whatever the market-power structure. As the continuous price increase allows the investors to obtain greater marginal profits than the landholders (i.e.,  $p-r > r-c_o$ ), the increases in the former's efforts and profits are higher than those of the latter. By comparing the effects of the two incentive solutions, we find that when the price rises, the profit gaps further increase between the two actors if the policy-makers only provide incentives to the investors. The profit gaps will undermine the fairness in REDD+, which is not conducive to achieving the neoliberal conservation.

Fig. 4 also reveals the influence of the market power on the actors' optimal strategies. Regardless of which incentive solution is active, the unequal market-power structure improves the efforts and profits of both landholders and investors. Moreover, whichever the incentive solutions, the gaps between the investors' efforts and profits under the condition of each market-power structure always increase as the price rises. Conversely, the effects of the market power on the landholders' efforts and profits emerge as more significant under the landholder-incentive solution.

### 3.2.5. Effects of the output elasticity of landholder efforts

Due to  $0 < \alpha < 1$  and  $0 < \alpha + \beta < 1$ , the output elasticity of landholder efforts ( $\alpha$ ) was set as 0.05–0.55. Fig. 5 shows the impacts of incentive solutions and market power on landholders and investors'

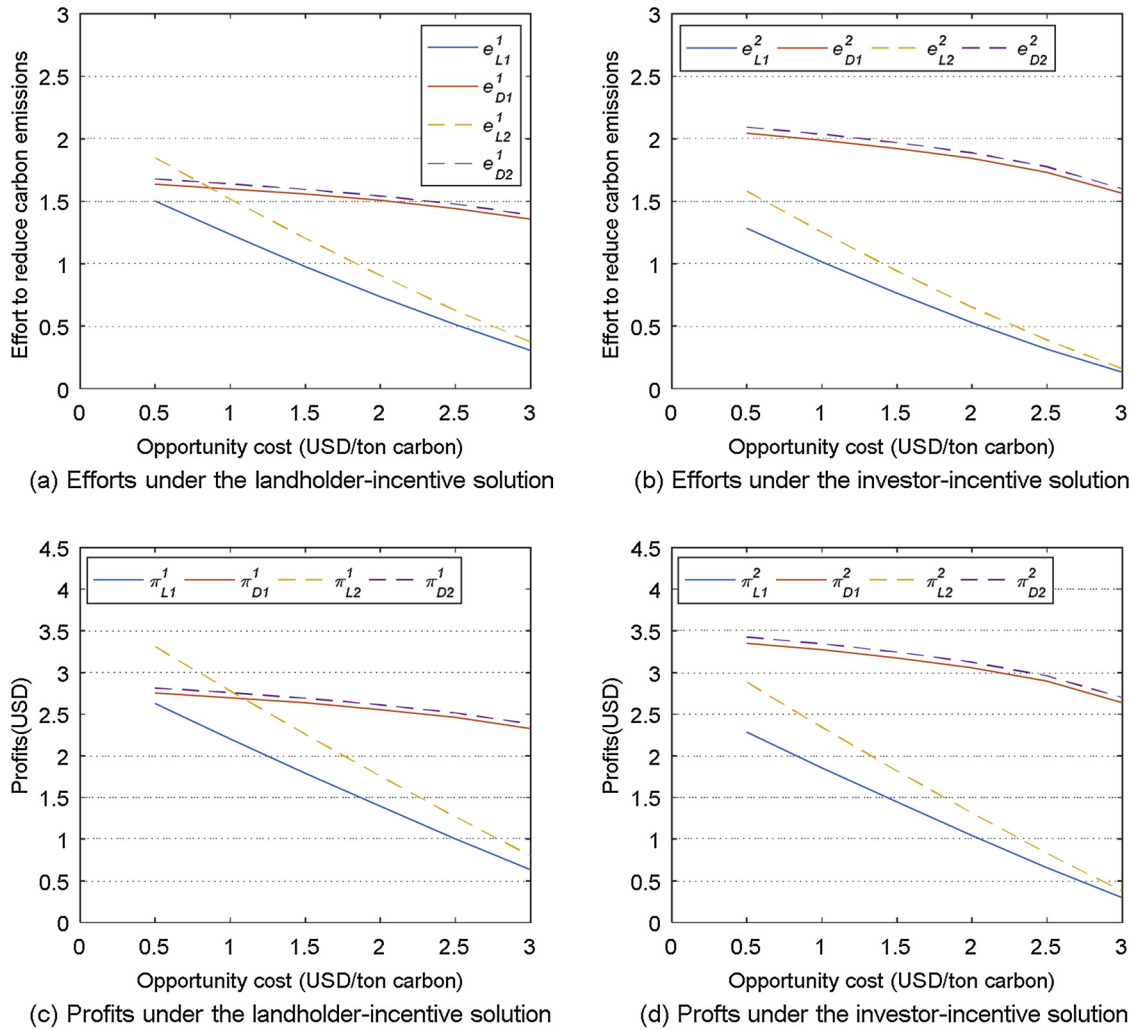


Fig. 3. The effects of opportunity cost per unit of reduced emissions ( $c_o$ ).

strategies when  $\alpha$  changes.

Fig. 5 shows the influences of the incentive solutions on the actors' optimal strategies. Whichever market-power structure is at play, both actors' efforts and profits are seen to rise as the output elasticity of landholder efforts ( $\alpha$ ) increases. However, the growth of the landholders' efforts and profit is shown to be faster than that of the investors. Furthermore, the efforts and profits of both actors under the landholder-incentive solution increase faster than those under the investor-incentive solution, particularly for the landholders. Landholders with high output elasticity can thus be seen to be more sensitive to the landholder-incentive solution. The implication here is that the latter solution could better encourage landholders to reduce emissions due to affording them more incentives than the other solution.

Fig. 5 also illustrates the role of the two market-power structures on actors' optimal strategies. Regardless of the incentive solution at work, the efforts and profits of both actors in the context of an unequal market-power structure can be seen to rise faster than those in an equal market-power structure. Moreover, the gaps between the two market-power structures increase as the output elasticity of landholder efforts ( $\alpha$ ) increases. These results also suggest that the influence of market-power structures on landholders is more significant than that on investors under the condition of either incentive solution.

### 3.2.6. Effects of the output elasticity of investor effort

Due to  $0 < \beta < 1$  and  $0 < \alpha + \beta < 1$ , the output elasticity of investor efforts ( $\beta$ ) was set as 0.05–0.55. Fig. 6 demonstrates the impacts of

incentive solutions and market power on landholders and investors' strategies when  $\beta$  changes.

Fig. 6 presents the effects of the incentive solutions on actors' behavior. Both actors' efforts and profits can be seen to rise as the output elasticity of investor efforts ( $\beta$ ) increases, especially for the investors. In contrast to the results in Fig. 5, the growth of the investors' efforts and profits emerges as faster than that of the landholders. Furthermore, the efforts and profits of both actors under the investor-incentive solution rise faster than those under the landholder-incentive solution, particularly for the investors. This outcome indicates that the investor-incentive solution is especially suitable for investors with high output elasticity since the latter is more sensitive to the investor-incentive solution than the landholder-incentive solution.

Fig. 6 also presents the effects of the two market-power structures on actors' behavior. Whichever incentive solution is active, the two actors' efforts and profits in the context of an unequal market-power structure are always shown to be higher than those in an equal market-power structure. These differences wrought by the two market-power structures increase as the output elasticity of landholder efforts ( $\alpha$ ) increases. Moreover, regardless of the incentive solution at play, landholders' efforts and profits are consistently subject to greater gaps under the two market-power structures than those of the investors. This suggests that the landholders are more sensitive to changes in the market-power structure as these actors play a leading role in REDD + . Thus, changes in the market power can be said to have more significant impacts on landholders than on investors as  $\beta$  continually rises.



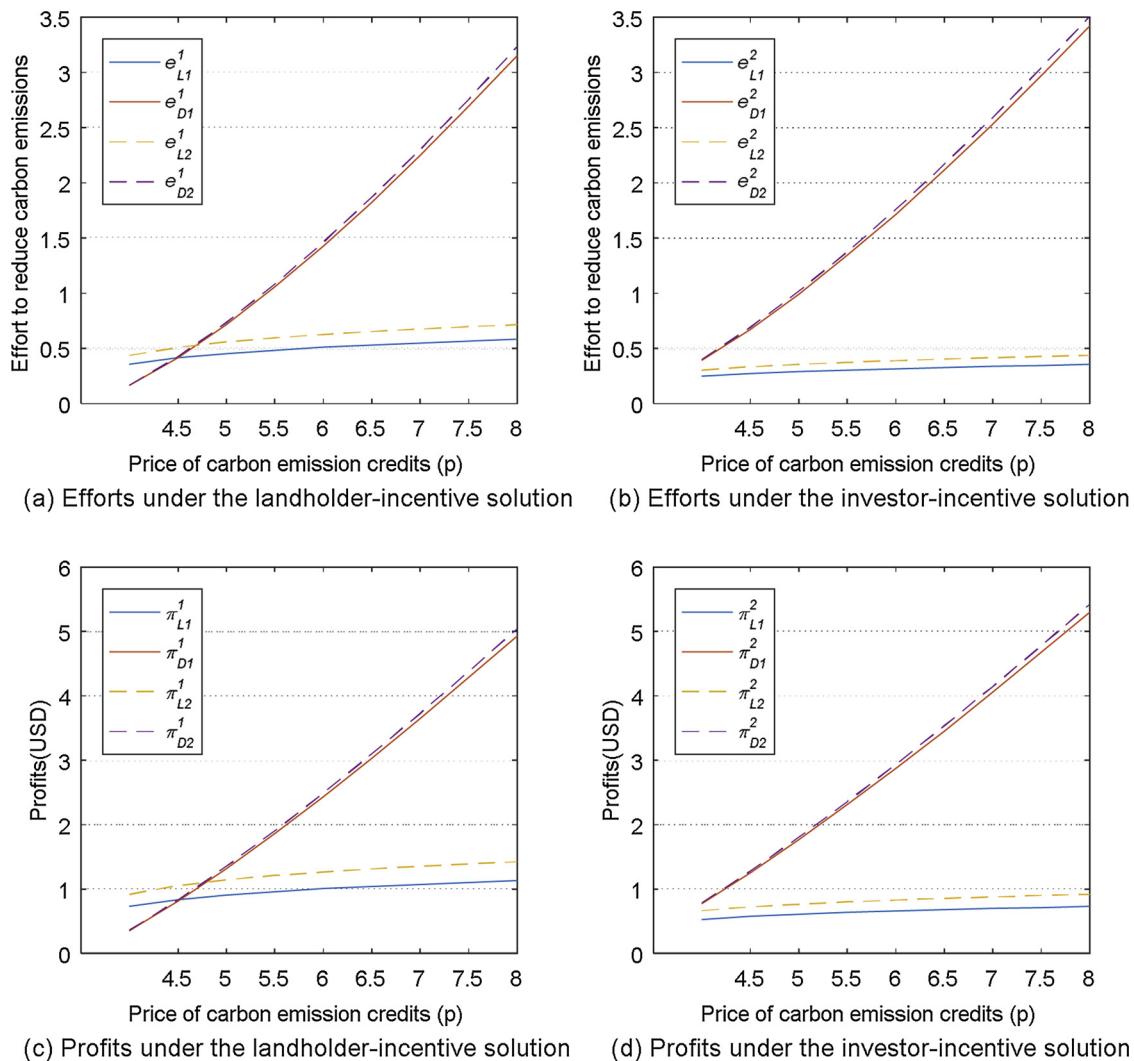


Fig. 4. The effects of the price of carbon emission credits ( $p$ ).

#### 4. Discussion

This paper investigates the impacts of incentive designs and market power on the conduct of key actors in the REDD + programme. The incentive designs aim to attract landholders and investors' sustained participation in the programme (Grosjean and Kontoleon, 2009). In deciding upon incentive solutions in the context of REDD +, the policy-makers first decide the motivating objects, while the other actors with differing degrees of market power then choose the optimal strategies to support their interests. Thus, the incentive designs together with the market power dynamics operating between landholders and investors, jointly shape REDD + outcomes.

Our study found that the incentives for landholders in REDD + are more conducive to affect actors' efforts to reduce emissions than the incentives for the investors. Specifically, the policy-makers use incentives to encourage people to avoid deforestation and to act in conservation friendly ways. Thus, REDD + incentives have become a key role in REDD + that constitutes a "regime of practices," guiding forest governance and conservation (McGregor et al., 2015). An important issue regarding REDD + incentive designs pertains to the question of who is eligible to obtain extra financial incentives from the policy-makers. This incentive design plays a vital role in the formulation of specific REDD + incentive policies, thereafter indirectly affecting actors' profits (Luttrell et al., 2013). Our results suggest that the landholder-incentive solution cannot only significantly increase

landholders' efforts and profits, but can also slightly improve investors' welfare. Since the incentives can offset external benefits, the landholders prefer to provide more ecosystem services. Meanwhile, involvement in ecosystem services represents a good deal for the investors if their marginal profits exceed marginal costs. The incentives will only motivate investors to participate in REDD + under the investor-incentive solution if the investors do not increase payments to the landholders. Due to not benefiting from the incentives, the landholders will not make additional efforts to reduce emissions. Thus, it can be argued that the investor-incentive solution is not an efficient or fair solution.

Landholders, rather than investors, can obtain more market power due to the unequal market-power structure in REDD +, which will ultimately improve the landholders' efforts to conserve forests and make a profit. Since the commodification of ecosystem services can reposition actors and create new social-economic hierarchies (Kronenberg and Hubacek, 2013), the REDD + scheme itself automatically creates market-power structures among its actors (Winkel, 2012). Such social-economic hierarchies often make the market power between landholders and investors unequal in REDD +, resulting in asymmetric decision-making sequences. Due to deciding whether to join REDD +, the landholders come to play a leading role, carrying the more market power that enables them to have priority in subsequent decision-making (Sheng et al., 2016). The results of our study suggest that unequal market-power structures significantly improve landholders' efforts and profits in REDD + due to the information asymmetry that

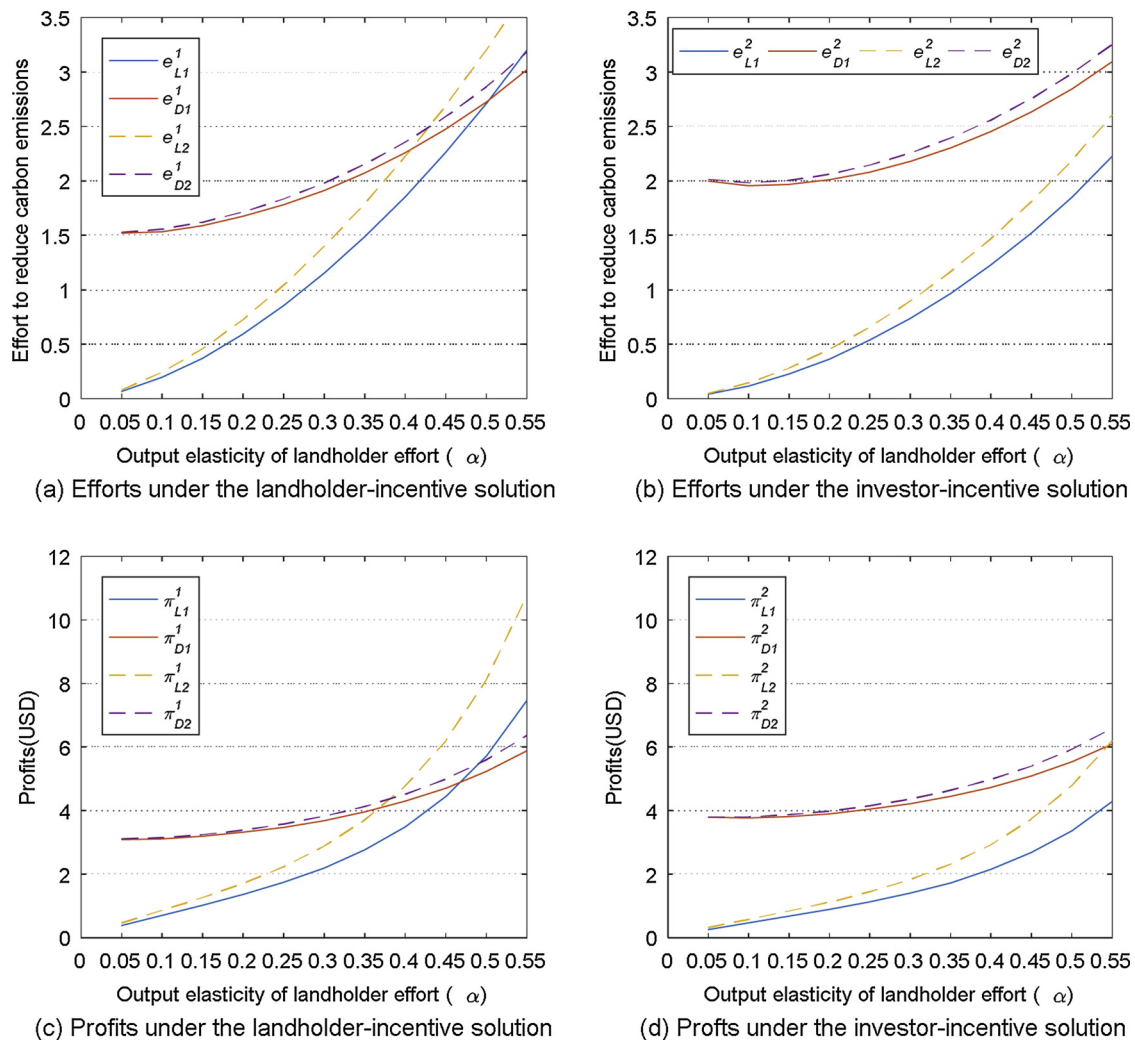


Fig. 5. The effects of output elasticity of landholder effort ( $\alpha$ ).

arises, to some extent (Kronenberg and Hubacek, 2013). More precisely, since the landholders are more aware of the opportunity costs of ecosystem services than the investors, this information asymmetry provides the landholders with more market power to extract informational rents (Ferraro, 2008). The landholders will thus make more efforts towards forest conservation in order to obtain more profits when these actors possess a greater degree of decision-making power.

However, our study also found that the landholders' advantages in market power will diminish with the increase in opportunity costs. Thus, joining REDD + will not be an optimal choice for the landholders when the opportunity costs of land-use change are very high (Irawan et al., 2013). Furthermore, information rents will not exist if the REDD + payments are lower than the opportunity costs (Karsenty et al., 2014). Since the investors as the follower can only make decisions based on the landholders' strategy when the market power is unequal, our study found no significant differences in changes to the investors' profits under either of the two market-power structures. The landholders' advantage in market power will grow as payments rise and, ultimately, undermine the investors' interests. However, the unequal market-power structure may also benefit the investors in the circumstance of an increase in the price of carbon credits. Moreover, the market-power inequality will increase the welfare of both actors when the output elasticity of landholders or investors' effort rises, which, in turn, induces them to make more efforts to reduce emissions. Thus, in the context of an unequal market-power structure, the actors with less market power can increase their profits by increasing the output

elasticity of their efforts.

Finally, incentive designs and market power have a greater impact on landholders than on investors. As our results show, the differences in profits caused by different market-power structures are even more pronounced when the policy-makers provide incentives to the landholders. When owning decision-making power in the landholder-incentive solution, the landholders are able to obtain more information rents and incentives, which can, in turn, encourage these actors to make more efforts in conserving forests and reducing emissions. As Brockhaus and Di Gregorio (2014) argued, "when the concentration of power is high, progress will be possible only if the interests of the most powerful are aligned with the objectives of REDD + and address the drivers of deforestation and forest degradation." Since investors' profits will be squeezed by the landholders' strategy if the investors are in a weak position in the decision-making process, the market power will exert little effect on the investors regardless of the incentive solution at play. Consequently, the investors will not increase their investment in REDD + even if they receive external incentives from the policy-makers. However, as the landholders will make more efforts to reduce emissions under the investor-incentive solution, in this circumstance the investors will be able to obtain more incentives from the policy-makers without extra efforts. Thus, empowering the landholders with both more decision-making power and more incentives can be seen as leading to potentially better outcomes in REDD +.

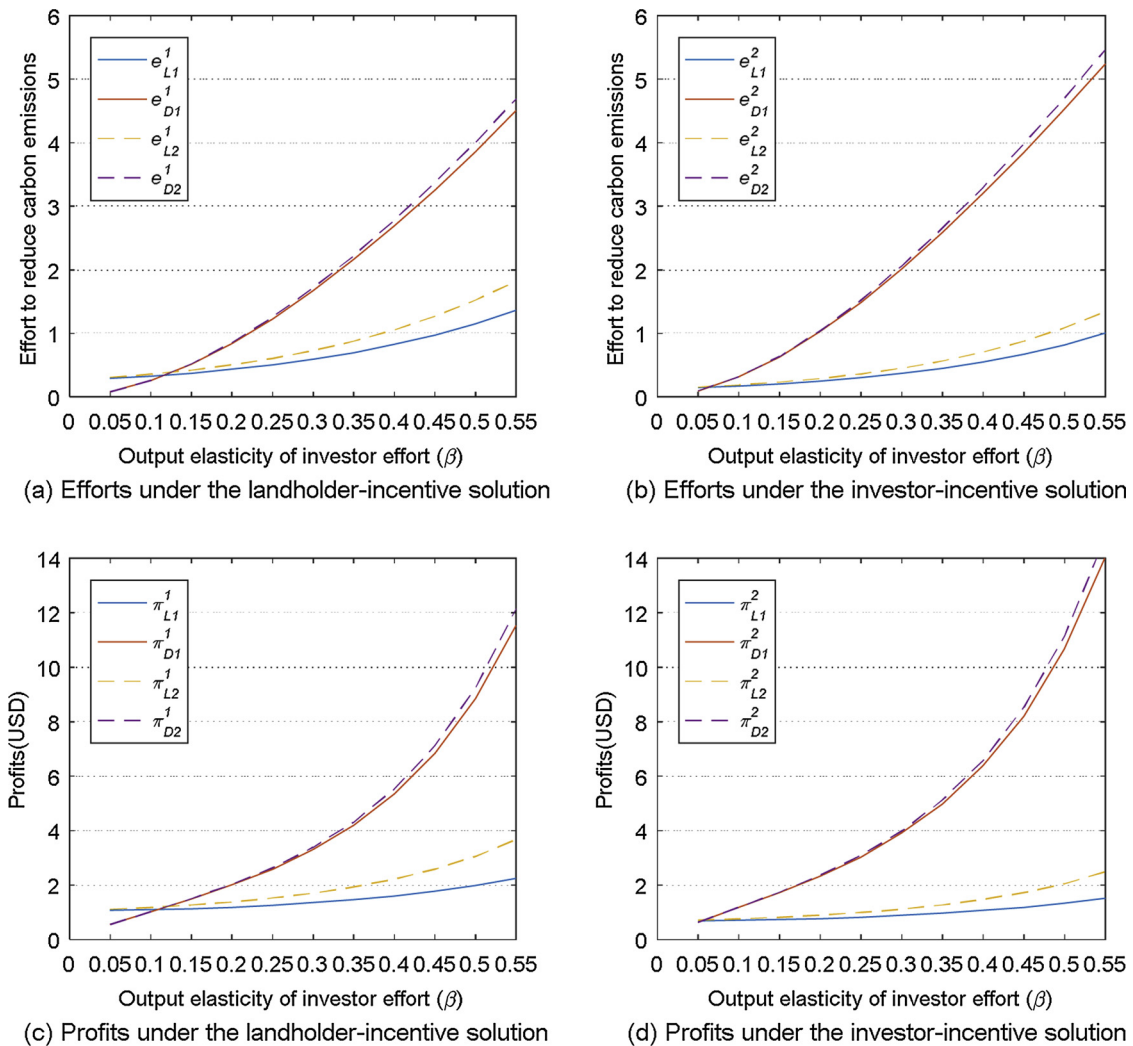


Fig. 6. The effects of output elasticity of investor effort ( $\beta$ ).

## 5. Conclusion

Market-power relationships are one of the key determinants of the success or failure of participatory natural resource management (Makatta et al., 2015). This resonates with the importance of considering market power in the context of REDD+ as the latter involves the multiple rights and interests of diverse actors. In addition, the incentive designs also play an important role in REDD+. REDD+, as an important application of neoliberalism in the conservation of natural resources, is essentially trying to make individuals acting as rational actors exhibit appropriate behavior by creating external incentives (Fletcher, 2010). As Youdelis (2013) argued, the most effective way to achieve conservation goals is often to marketize nature and establish a monetary incentive structure to motivate rational actors. Within a theoretical framework that considers both market power and incentive designs, this paper examines the impacts of incentive designs and market power on actors' behavior in REDD+ by constructing several game models.

Indeed, different incentive designs have a heterogeneous impact on actors' behavior. The difference in market power between actors makes this heterogeneous impact more complicated. At present, a large amount of literature mainly focuses on the changes in the behavior of various actors in REDD+ by observing specific case studies (such as McAfee and Shapiro (2010); Astuti and McGregor (2015); McGregor et al. (2015), and Boer (2017)). Compared with the existing political and economic studies on REDD+, this paper adds a new understanding

of the role of incentive designs and market power in the heterogeneity of REDD+. Our contribution can be summarized in three components. First, we construct a theoretical framework that considers market power differences and different incentive solutions simultaneously, to reveal the roles of the market power and incentive designs in REDD+. Second, we establish game models and use numerical simulation to provide a feasible, quantitative approach for studying the role of market-power relationships and incentive solutions in REDD+. Third, we reveal the mechanism by which market power and incentive designs affect the actors' behavior. The results demonstrate that the landholder-incentive solution can increase the efforts and profits of the landholders and investors simultaneously. However, the investor-incentive solution only enhances investors' profits when the payments are fixed. Since unequal market power can encourage actors to make more efforts to reduce emissions, landholders with market power can benefit from this structure, and investors' efforts and profits in this market-power context will also not decline.

Furthermore, our findings show that the influence of market power and incentive designs on actors also depends on several factors, including opportunity costs, payments, the price of carbon credits, and output elasticities of effort. Thus, the actors with less market power can improve their profits by increasing their output elasticity of effort when the market power is unequal. Moreover, incentive designs and market power have a greater impact on landholders than on investors. In summary, differences in market power and motivating objects not only affect the distribution of interests among actors but also influence the

effectiveness of REDD + policies.

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## Appendix A

Optimal solutions under the landholder-incentive solutions

### A.1 Solutions in an equal market-power structure

When the landholders and investors have equal market power, they can concurrently make decisions to maximize their profits. Thus, their profit functions are as follows:

$$\begin{cases} \max_{e_{L1}} \pi_{L1}^1 = (r - co + s)Ae_{L1}^\alpha e_{D1}^\beta - \frac{\theta e_{L1}^2}{2} \\ \max_{e_{D1}} \pi_{D1}^1 = (p - r)Ae_{L1}^\alpha e_{D1}^\beta - \frac{\theta e_{D1}^2}{2} \end{cases} \quad (A.1.1)$$

Where  $\pi_{L1}^1$  and  $\pi_{D1}^1$  represent the profits of landholders and investors with equal internal power under the landholder-incentive solution, respectively. Thus, the first- and second-order derivatives of the two profit functions are as follows:

$$\begin{cases} \frac{\partial \pi_{L1}^1}{\partial e_{L1}} = (r - co + s)\alpha Ae_{L1}^{\alpha-1} e_{D1}^\beta - \theta e_{L1} \\ \frac{\partial \pi_{D1}^1}{\partial e_{D1}} = (p - r)\beta Ae_{L1}^\alpha e_{D1}^{\beta-1} - \theta e_{D1} \end{cases} \quad (A.1.2)$$

and

$$\begin{cases} \frac{\partial^2 \pi_{L1}^1}{\partial e_{L1}^2} = (r - co + s)\alpha(\alpha - 1)Ae_{L1}^{\alpha-2} e_{D1}^\beta - \theta < 0 \\ \frac{\partial^2 \pi_{D1}^1}{\partial e_{D1}^2} = (p - r)\beta(\beta - 1)Ae_{L1}^\alpha e_{D1}^{\beta-2} - \theta < 0 \end{cases} \quad (A.1.3)$$

Due to  $p > r > c_o > 0$ ,  $0 < \alpha < 1$  and  $0 < \beta < 1$ , the second-order derivative of each profit function is negative, which suggests that each profit function is strictly concave. Thus, each profit function has a maximum. By setting the first-order derivative of each profit function to zero, the optimal efforts of landholders and investors ( $e_{L1}^1$  and  $e_{D1}^1$ ) are as follows:

$$\begin{cases} e_{L1}^1 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p-r))^{\frac{\beta}{2(2-\alpha-\beta)}} (\alpha(r-co+s))^{\frac{2-\beta}{2(2-\alpha-\beta)}} \\ e_{D1}^1 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p-r))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} (\alpha(r-co+s))^{\frac{\alpha}{2(2-\alpha-\beta)}} \end{cases} \quad (A.1.4)$$

Substituting Eq. (A.1.4) into Eq. (A.1.1), the optimal profits of landholders and investors with equal market power under the landholder-incentive solution are as follows:

$$\begin{cases} \pi_{L1}^1 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r-co+s))^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p-r))^{\frac{\beta}{2-\alpha-\beta}} \left(\alpha^{\frac{\alpha}{2-\alpha-\beta}} - \frac{1}{2}\right) \\ \pi_{D1}^1 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r-co+s))^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p-r))^{\frac{2-\alpha}{2-\alpha-\beta}} \left(\beta^{\frac{\beta}{2-\alpha-\beta}} - \frac{1}{2}\right) \end{cases} \quad (A.1.5)$$

### A.2 Solutions in an unequal market-power structure

The landholders and investors make decisions, in turn, to maximize their profits in an unequal market-power structure. Thus, their profit functions are:

$$\max_{e_{L2}} \pi_{L2}^1 = (r - co + s)Ae_{L2}^\alpha e_{D2}^\beta - \frac{\theta e_{L2}^2}{2} \quad (A.2.1)$$



$$\max_{e_{D2}} \pi_{D2}^1 = (p - r)Ae_{L2}^\alpha e_{D2}^\beta - \frac{\theta e_{D2}^2}{2} \quad (\text{A.2.2})$$

Where  $\pi_{L2}^1$  and  $\pi_{D2}^1$  represent the profits of landholders and investors with unequal market power under the landholder-incentive solution, respectively. Since the landholders are the leaders in the game model, the backward induction is applied to solve it. Setting the first-order derivative of the investor profit function to be zero:

$$\frac{\partial \pi_{D2}^1}{\partial e_{D2}} = (p - r)\beta Ae_{L2}^\alpha e_{D2}^{\beta-1} - \theta e_{D2} = 0 \quad (\text{A.2.3})$$

Thus, the investors' best response function is as follows:

$$e_{D2}^* = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\beta}} (\beta(p - r))^{\frac{1}{2-\beta}} e_{L2}^{\frac{\alpha}{2-\beta}} \quad (\text{A.2.4})$$

Substituting Eq. (A.2.4) into Eq. (A.2.1), the landholders' profit function is:

$$\pi_{L2}^1 = (r - co + s)A^{\frac{2}{2-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\beta}{2-\beta}} (\beta(p - r))^{\frac{\beta}{2-\beta}} e_{L2}^{\frac{2\alpha}{2-\beta}} - \frac{\theta e_{L2}^2}{2} \quad (\text{A.2.5})$$

The first- and second-order derivatives of the landholders' profit function are:

$$\frac{\partial \pi_{L2}^1}{\partial e_{L2}} = (r - co + s)A^{\frac{2}{2-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\beta}{2-\beta}} (\beta(p - r))^{\frac{\beta}{2-\beta}} \frac{2\alpha}{2-\beta} e_{L2}^{\frac{2\alpha}{2-\beta}-1} - \theta e_{L2} \quad (\text{A.2.6})$$

and

$$\frac{\partial^2 \pi_{L2}^1}{\partial e_{L2}^2} = (r - co + s)A^{\frac{2}{2-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\beta}{2-\beta}} (\beta(p - r))^{\frac{\beta}{2-\beta}} \frac{2\alpha}{2-\beta} \left( \frac{2\alpha}{2-\beta} - 1 \right) e_{L2}^{\frac{2\alpha}{2-\beta}-2} - \theta < 0 \quad (\text{A.2.7})$$

Due to  $p > r > c_o > 0$ ,  $0 < \alpha < 1$  and  $0 < \beta < 1$ , the second-order derivative (i.e. Eq. (A.2.7)) is negative. It suggests that the landholder profit function is a strictly concave function, therefore, owning a maximum. By setting the first-order derivative (i.e. Eq. (A.2.6)) be zero, the landholders' optimal efforts are as follows:

$$e_{L2}^1 = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\alpha-\beta}} \left( \frac{2\alpha}{2-\beta} \right)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (r - co + s)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (\beta(p - r))^{\frac{\beta}{2(2-\alpha-\beta)}} \quad (\text{A.2.8})$$

Substituting Eq. (A.2.8) into Eq. (A.2.4), the investors' optimal efforts are as follows:

$$e_{D2}^1 = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\alpha-\beta}} \left( \frac{2\alpha}{2-\beta} \right)^{\frac{\alpha}{2(2-\alpha-\beta)}} (r - co + s)^{\frac{\alpha}{2(2-\alpha-\beta)}} (\beta(p - r))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} \quad (\text{A.2.9})$$

Finally, the optimal profits of landholders and investors with unequal market power under the landholder-incentive solution are as follows by substituting Eq. (A.2.8) and (A.2.9) into Eq. (A.2.1) and (A.2.2), respectively:

$$\left\{ \begin{array}{l} \pi_{L2}^1 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left( \frac{2\alpha(r - co + s)}{2-\beta} \right)^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p - r))^{\frac{\beta}{2-\alpha-\beta}} \\ \quad \left( \left( \frac{2\alpha}{2-\beta} \right)^{\frac{\alpha}{2-\beta}} - \frac{1}{2} \right) \\ \pi_{D2}^1 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left( \frac{2\alpha(r - co + s)}{2-\beta} \right)^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p - r))^{\frac{2-\alpha}{2-\alpha-\beta}} \left( \beta^{\frac{\beta}{2-\alpha}} \right. \\ \quad \left. - \frac{1}{2} \right) \end{array} \right. \quad (\text{A.2.10})$$

## Appendix B

### Optimal solutions under the investor-incentive solutions

#### B.1 Solutions in an equal market-power structure

Like the situation under the landholder-incentive solution, the landholders and investors under the investor-incentive solution can concurrently make decisions when they have equal market power. Thus, their profit functions are as follows:

$$\left\{ \begin{array}{l} \max_{e_{L1}} \pi_{L1}^2 = (r - co)Ae_{L1}^\alpha e_{D1}^\beta - \frac{\theta e_{L1}^2}{2} \\ \max_{e_{D1}} \pi_{D1}^2 = (p - r + s)Ae_{L1}^\alpha e_{D1}^\beta - \frac{\theta e_{D1}^2}{2} \end{array} \right. \quad (\text{B.1.1})$$

Where  $\pi_{L1}^2$  and  $\pi_{D1}^2$  represent the profits of landholders and investors with equal market power under the investor-incentive solution, respectively. Thus, the first- and second-order derivatives of the two profit functions are as follows:

$$\begin{cases} \frac{\partial \pi_{L1}^2}{\partial e_{L1}} = (r - co)\alpha A e_{L1}^{\alpha-1} e_{D1}^\beta - \theta e_{L1} \\ \frac{\partial \pi_{D1}^2}{\partial e_{D1}} = (p - r + s)\beta A e_{L1}^\alpha e_{D1}^{\beta-1} - \theta e_{D1} \end{cases} \quad (B.1.2)$$

and

$$\begin{cases} \frac{\partial^2 \pi_{L1}^2}{\partial e_{L1}^2} = (r - co)\alpha(\alpha - 1)A e_{L1}^{\alpha-2} e_{D1}^\beta - \theta < 0 \\ \frac{\partial^2 \pi_{D1}^2}{\partial e_{D1}^2} = (p - r + s)\beta(\beta - 1)A e_{L1}^\alpha e_{D1}^{\beta-2} - \theta < 0 \end{cases} \quad (B.1.3)$$

Due to  $p > r > c_o > 0$ ,  $0 < \alpha < 1$  and  $0 < \beta < 1$ , the second-order derivative of each profit function is negative. Thus, each profit function has a maximum. By setting the first-order derivative of each profit function to zero, the optimal efforts of landholders and investors ( $e_{L1}^2$  and  $e_{D1}^2$ ) are as follows:

$$\begin{cases} e_{L1}^2 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p - r + s))^{\frac{\beta}{2(2-\alpha-\beta)}} (\alpha(r - co))^{\frac{2-\beta}{2(2-\alpha-\beta)}} \\ e_{D1}^2 = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\alpha-\beta}} (\beta(p - r + s))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} (\alpha(r - co))^{\frac{\alpha}{2(2-\alpha-\beta)}} \end{cases} \quad (B.1.4)$$

Substituting Eq. (B.1.4) into Eq. (B.1.1), the optimal profits of landholders and investors with equal market power under the investor-incentive solution are as follows:

$$\begin{cases} \pi_{L1}^2 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r - co))^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p - r + s))^{\frac{\beta}{2-\alpha-\beta}} \left(\alpha^{\frac{\alpha}{2-\beta}} - \frac{1}{2}\right) \\ \pi_{D1}^2 = A^{\frac{2}{2-\alpha-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} (\alpha(r - co))^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p - r + s))^{\frac{2-\alpha}{2-\alpha-\beta}} \left(\beta^{\frac{\beta}{2-\alpha}} - \frac{1}{2}\right) \end{cases} \quad (B.1.5)$$

## B.2 Solutions in an unequal market-power structure

Like the situation under the landholder-incentive solution, the landholders and investors under the investor-incentive solution will make decisions in turn in an unequal market-power structure. Thus, their profit functions are:

$$\max_{e_{L2}} \pi_{L2}^2 = (r - co)A e_{L2}^\alpha e_{D2}^\beta - \frac{\theta e_{L2}^2}{2} \quad (B.2.1)$$

$$\max_{e_{D2}} \pi_{D2}^2 = (p - r + s)A e_{L2}^\alpha e_{D2}^\beta - \frac{\theta e_{D2}^2}{2} \quad (B.2.2)$$

Where  $\pi_{L2}^2$  and  $\pi_{D2}^2$  represent the profits of landholders and investors with unequal market power under the investor-incentive solution, respectively. Setting the first-order derivative of the investor profit function to be zero according to the backward induction:

$$\frac{\partial \pi_{D2}^2}{\partial e_{D2}} = (p - r + s)\beta A e_{L2}^\alpha e_{D2}^{\beta-1} - \theta e_{D2} = 0 \quad (B.2.3)$$

The investors' best response function can be obtained as follows:

$$e_{D2}^* = \left(\frac{A}{\theta}\right)^{\frac{1}{2-\beta}} (\beta(p - r + s))^{\frac{1}{2-\beta}} e_{L2}^{\frac{\alpha}{2-\beta}} \quad (B.2.4)$$

Similarly, substituting Eq. (B.2.4) into Eq. (B.2.1), the landholders' profit function is:

$$\pi_{L2}^2 = (r - co)A^{\frac{2}{2-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\beta}{2-\beta}} (\beta(p - r + s))^{\frac{\beta}{2-\beta}} e_{L2}^{\frac{2\alpha}{2-\beta}} - \frac{\theta e_{L2}^2}{2} \quad (B.2.5)$$

The first- and second-order derivatives of the landholders' profit function are:

$$\frac{\partial \pi_{L2}^2}{\partial e_{L2}} = (r - co)A^{\frac{2}{2-\beta}} \left(\frac{1}{\theta}\right)^{\frac{\beta}{2-\beta}} (\beta(p - r + s))^{\frac{\beta}{2-\beta}} \frac{2\alpha}{2-\beta} e_{L2}^{\frac{2\alpha}{2-\beta}-1} - \theta e_{L2} \quad (B.2.6)$$

and

$$\frac{\partial^2 \pi_{L2}^2}{\partial e_{L2}^2} = (r - co) A^{\frac{2}{2-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\beta}{2-\beta}} (\beta(p - r + s))^{\frac{\beta}{2-\beta}} \frac{2\alpha}{2-\beta} \left( \frac{2\alpha}{2-\beta} - 1 \right) e_{L2}^{\frac{2\alpha}{2-\beta}-2} - \theta < 0 \quad (\text{B.2.7})$$

Due to  $p > r > c_o > 0$ ,  $0 < \alpha < 1$  and  $0 < \beta < 1$ , the second-order derivative (i.e. Eq. (B.2.7)) is negative. Thus, landholders have a maximal profit. By setting the first-order derivative (i.e. Eq. (B.2.6)) be zero, the landholders' optimal efforts are as follows:

$$e_{L2}^2 = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\alpha-\beta}} \left( \frac{2\alpha}{2-\beta} \right)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (r - co)^{\frac{2-\beta}{2(2-\alpha-\beta)}} (\beta(p - r + s))^{\frac{\beta}{2(2-\alpha-\beta)}} \quad (\text{B.2.8})$$

Substituting Eq. (B.2.8) into Eq. (B.2.4), the investors' optimal efforts are as follows:

$$e_{D2}^2 = \left( \frac{A}{\theta} \right)^{\frac{1}{2-\alpha-\beta}} \left( \frac{2\alpha}{2-\beta} \right)^{\frac{\alpha}{2(2-\alpha-\beta)}} (r - co)^{\frac{\alpha}{2(2-\alpha-\beta)}} (\beta(p - r + s))^{\frac{2-\alpha}{2(2-\alpha-\beta)}} \quad (\text{B.2.9})$$

Finally, the optimal profits of landholders and investors with unequal market power under the investor-incentive solution are as follows by substituting Eq. (B.2.8) and (B.2.9) into Eq. (B.2.1) and (B.2.2), respectively:

$$\begin{cases} \pi_{L2}^2 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left( \frac{2\alpha(r - co)}{2-\beta} \right)^{\frac{2-\beta}{2-\alpha-\beta}} (\beta(p - r + s))^{\frac{\beta}{2-\alpha-\beta}} \\ \quad \left( \left( \frac{2\alpha}{2-\beta} \right)^{\frac{\alpha}{2-\beta}} - \frac{1}{2} \right) \\ \pi_{D2}^2 = A^{\frac{2}{2-\alpha-\beta}} \left( \frac{1}{\theta} \right)^{\frac{\alpha+\beta}{2-\alpha-\beta}} \left( \frac{2\alpha(r - co)}{2-\beta} \right)^{\frac{\alpha}{2-\alpha-\beta}} (\beta(p - r + s))^{\frac{2-\alpha}{2-\alpha-\beta}} \left( \frac{\beta}{2-\alpha} \right. \\ \quad \left. - \frac{1}{2} \right) \end{cases} \quad (\text{B.2.10})$$

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